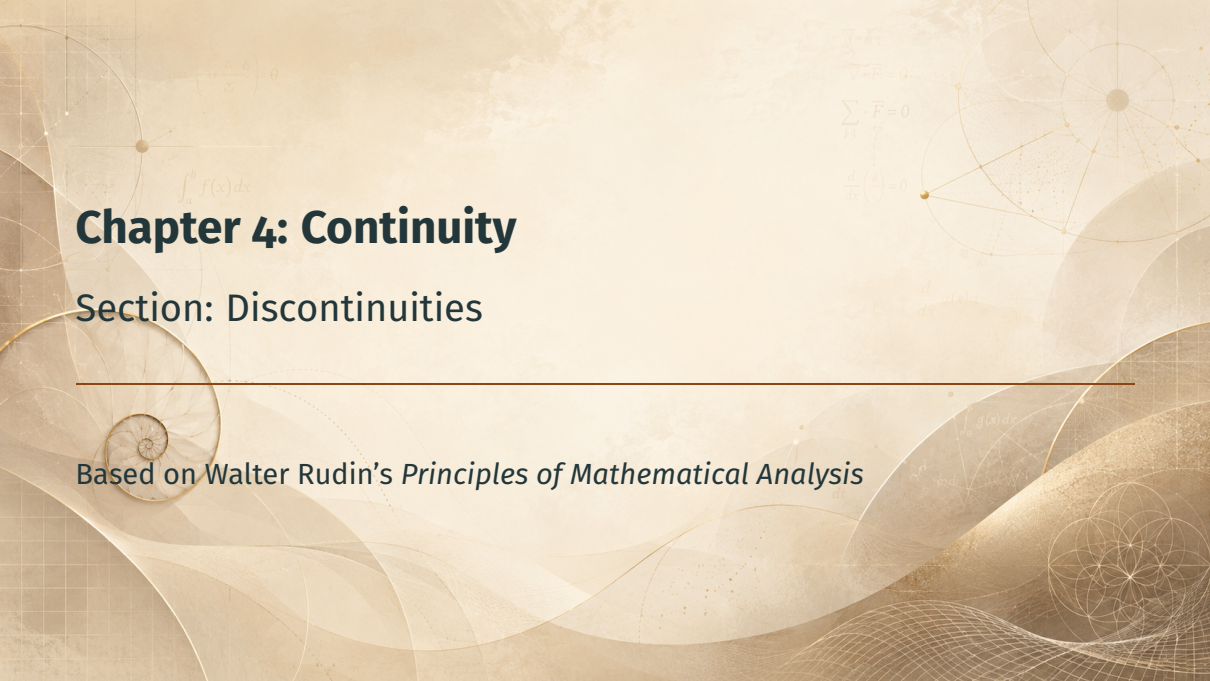


The background is a light beige color with various mathematical and geometric elements. In the top left, there is a formula $\left(\frac{\partial \phi}{\partial x} \right) = 0$. Below it is $\int_a^b f(x) dx$. In the top right, there are formulas $\nabla \cdot \vec{F} = 0$, $\sum_{i=1}^n \vec{F}_i = 0$, and $\frac{d}{dx} \left(\frac{g}{f} \right) = 0$. On the right side, there is a diagram of a sphere with lines connecting points on its surface. In the bottom right, there is a diagram of a circle with internal lines forming a star-like pattern. A horizontal line is drawn across the middle of the page.

Chapter 4: Continuity

Section: Discontinuities

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

One-Sided Limits

Two Kinds of Discontinuity

Examples

Summary

Motivation

Why Classify Discontinuities?

- We have studied where functions **are** continuous

Why Classify Discontinuities?

- We have studied where functions **are** continuous
- Now: what happens where a function **fails** to be continuous?

Why Classify Discontinuities?

- We have studied where functions **are** continuous
- Now: what happens where a function **fails** to be continuous?
- Goal: sort discontinuities into **simple** vs. **wild**

Recall: Continuity at a Point

Definition 4.5 (recap)

f is **continuous at x** if $f(t) \rightarrow f(x)$ as $t \rightarrow x$; equivalently $\lim_{t \rightarrow x} f(t) = f(x)$.

So f is *discontinuous* at x if this fails: the limit disagrees with $f(x)$, or the limit does not exist at all.

$$\nabla \cdot F = 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\int_a^b f(x) dx$$

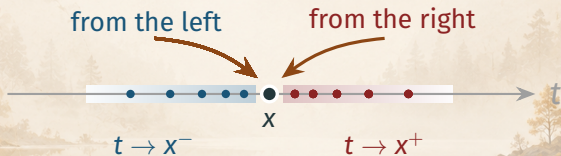
$$e^{i\pi} + 1 = 0$$

One-Sided Limits

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The Idea of One-Sided Limits

Approaching x from the **right** and from the **left** may give different answers.



Notation: $f(x+)$ is the **right-hand limit**, $f(x-)$ the **left-hand limit**.

Definition 4.25: Right- and Left-Hand Limits

Definition 4.25

Let f be defined on (a, b) . For $a \leq x < b$, write

$$f(x+) = q$$

if $f(t_n) \rightarrow q$ for **every** sequence $\{t_n\}$ in (x, b) with $t_n \rightarrow x$.

For $a < x \leq b$, define $f(x-)$ using sequences $\{t_n\}$ in (a, x) .

When Does the Full Limit Exist?

Key observation

For any $x \in (a, b)$, $\lim_{t \rightarrow x} f(t)$ exists **if and only if**

$$f(x+) = f(x-) = \lim_{t \rightarrow x} f(t).$$

The two-sided limit exists iff both one-sided limits exist *and agree*.

$$e^{i\pi} + 1 = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

Two Kinds of Discontinuity

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Definition 4.26: First and Second Kind

Definition 4.26

Suppose f is discontinuous at x .

- If **both** $f(x+)$ and $f(x-)$ exist: discontinuity of the **first kind**, or a **simple discontinuity**.
- Otherwise: discontinuity of the **second kind**.

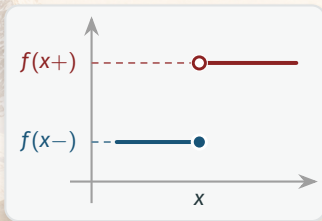
The dividing line: do both one-sided limits exist?

Two Flavors of Simple Discontinuity

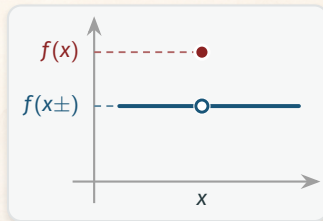
Both $f(x+)$ and $f(x-)$ exist, yet f is still discontinuous. Two ways:

(1) $f(x+) \neq f(x-)$ — a **jump** (the value $f(x)$ is irrelevant).

(2) $f(x+) = f(x-) \neq f(x)$ — a **removable** discontinuity.

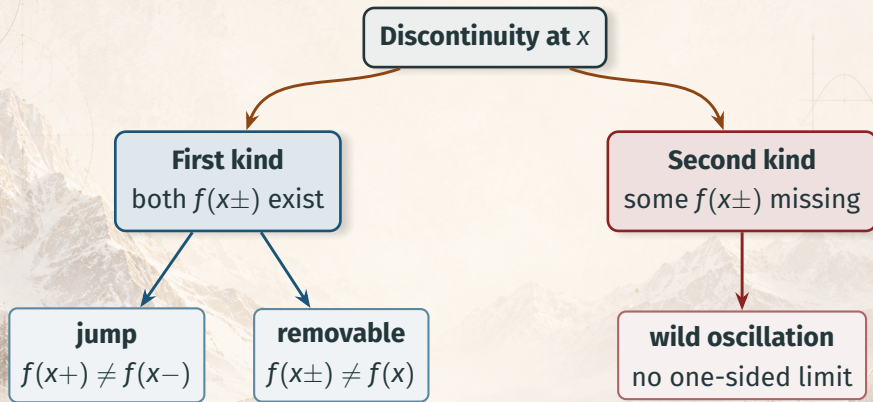


(1) jump



(2) removable

The Full Taxonomy



Examples

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\nabla^2 \phi = 0$$

$$e^{im} + 1 = 0$$

$$\frac{\partial u}{\partial t} = \sigma \nabla^2 u$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Example 4.27(a): The Dirichlet Function

Example 4.27(a)

$$f(x) = \begin{cases} 1 & (x \text{ rational}), \\ 0 & (x \text{ irrational}). \end{cases}$$

At **every** point, neither $f(x+)$ nor $f(x-)$ exists $\Rightarrow$ discontinuity of the **second kind** everywhere.

Example 4.27(b): A Single Point of Rescue

Example 4.27(b)

$$f(x) = \begin{cases} x & (x \text{ rational}), \\ 0 & (x \text{ irrational}). \end{cases}$$

Continuous at $x = 0$; discontinuity of the **second kind** at every other point.

Why $x = 0$ **works**: near 0, both branches (x and 0) are tiny, so $f(t) \rightarrow 0 = f(0)$.

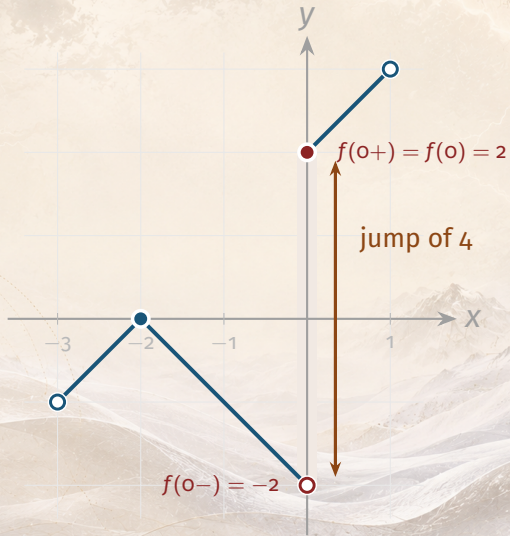
Example 4.27(c): A Genuine Jump

Example 4.27(c)

$$f(x) = \begin{cases} x + 2 & (-3 < x < -2), \\ -x - 2 & (-2 \leq x < 0), \\ x + 2 & (0 \leq x < 1). \end{cases}$$

Simple discontinuity at $x = 0$; continuous elsewhere on $(-3, 1)$.

Example 4.27(c): The Graph



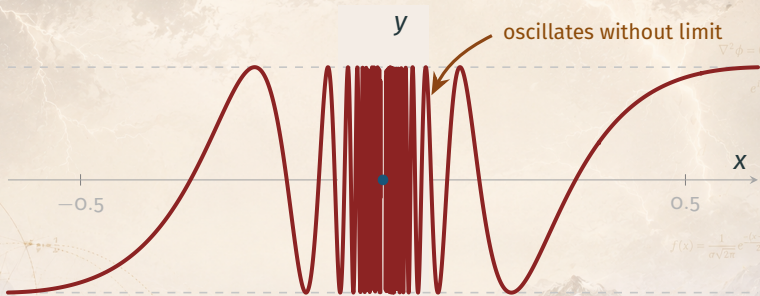
Example 4.27(d): Wild Oscillation

Example 4.27(d)

$$f(x) = \begin{cases} \sin \frac{1}{x} & (x \neq 0), \\ 0 & (x = 0). \end{cases}$$

Neither $f(0+)$ nor $f(0-)$ exists $\Rightarrow$ discontinuity of the **second kind** at 0; continuous for $x \neq 0$.

Example 4.27(d): The Graph




$$e^{i\pi} + 1 = 0$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

Summary

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- **One-sided limits** $f(x+)$, $f(x-)$: approach x from one side.

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways:

- **One-sided limits** $f(x+)$, $f(x-)$: approach x from one side.
- **First kind** (simple): both one-sided limits exist — a jump or a removable hole.

Summary

Key takeaways:

- **One-sided limits** $f(x+)$, $f(x-)$: approach x from one side.
- **First kind** (simple): both one-sided limits exist — a jump or a removable hole.
- **Second kind**: at least one one-sided limit fails to exist.

Summary

Key takeaways:

- **One-sided limits** $f(x+)$, $f(x-)$: approach x from one side.
- **First kind** (simple): both one-sided limits exist — a jump or a removable hole.
- **Second kind**: at least one one-sided limit fails to exist.

Coming up next: *Monotonic Functions* — where simple discontinuities are the **only** kind possible.